

Bing

Let's prove that $F \cap G$ is a sigma-field on Ω .

To do so, we need to show that:

1. $\Omega \in F \cap G$
2. If $A \in F \cap G$, then $A^c \in F \cap G$
3. If $A_1, A_2, \dots \in F \cap G$, then $\bigcup_{n=1}^{\infty} A_n \in F \cap G$
4. Since F and G are sigma-fields on Ω , we know that $\Omega \in F$ and $\Omega \in G$.
Therefore, $\Omega \in F \cap G$.
5. Let $A \in F \cap G$. Then $A \in F$ and $A \in G$. Since F and G are sigma-fields on Ω , we know that $A^c \in F$ and $A^c \in G$. Therefore, $A^c \in F \cap G$.
6. Let $A_1, A_2, \dots \in F \cap G$. Then $A_i \in F$ and $A_i \in G$ for all i . Since F and G are sigma-fields on Ω , we know that $\bigcup_{n=1}^{\infty} A_i \in F$ and $\bigcup_{n=1}^{\infty} A_i \in G$.
Therefore, $\bigcup_{n=1}^{\infty} A_i \in F \cap G$.

Therefore, we have shown that $F \cap G$ is a sigma-field on Ω .

Now let's show an example where $F \cup G$ fails to be a sigma-field on Ω .

Let $\Omega = \{a, b, c\}$, let $F = \{\{a, b\}, \{c\}, \{a, c\}, \{\emptyset\}\}$, and let $G = \{\{a\}, \{b, c\}, \{a, b, c\}, \{\emptyset\}\}$.

Then we have:

$$F \cup G = \{\{a, b\}, \{c\}, \{a, c\}, \{\emptyset\}, \{a\}, \{b, c\}, \{a, b, c\}\}$$

However, $\{b, c\}$ is not in $F \cup G$ since it is not in either of the sets in the union.

Therefore, we have shown that $F \cup G$ may fail to be a sigma-field on Ω .